SdS

→ Loss functions in Deep learning

- Loss function is a Method of evaluating how well your algorithm is Modelling your dataset

- High value → poor performance of Model
- Low value → great " "

↳ we can change our loss fn with the help of changing in the parameter of algorithm.

Why important?

↳ you can't improve what you can't measure
↳ Peter Drucker

→ Type of Loss fn in DL

* Regression
  - → MSE
  - → MAE
  - → Huber loss
* Classification
  - → Binary Cross entropy
  - → Hinge loss
  - → Categorical cross entropy
* Autoencoders
  - → KL divergence
* GAN
  - → discriminator loss
  - → MinMax gan loss
* Object detection
  - → focal loss
* Embedding
  - → Tripled loss

ShivamCoder

SRS

Loss fn → single training eg: $(y_i - \hat{y}_i)^2$

Cost fn → for batch data / whole data eg: $\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$

1) Mean Squared Error (MSE) / L2 Loss

$$MSE = (y_i - \hat{y}_i)^2$$

(true value − predicted value)²

Main goal → to minimize the loss fn value

Why square → to handle negative value → if we not handle then it decrease our loss fn eg: +4 −2 +4 −2 ⇒ +4 ✗

$(+4)^2 + (-2)^2 + (+4)^2 + (-2)^2 \Rightarrow 16 + 4 + 16 + 4 \Rightarrow 40$

- quadratic fn due to that for point value changes big drastic change in our model

Advantages

1) Easy to interpret
2) Always Differentiable
3) 1 Local Minima

Disadvantage

1) Error unit (square) eg: lpa → $(lpa)^2$
2) Not Robust to outliers ✗
3) Activation fn → linear for last neuron (output)

2) Mean Absolute Error (MAE) → L1 Loss

$$L = |y_i - \hat{y}_i|$$

$$C = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

SDS

Advantage

1) Intuitive and easy
2) Unit → same → y
3) Robust to outliers

Disadvantage

1) Not differentiable

3) Huber Loss

$$L = \begin{cases} \frac{1}{2}(y-\hat{y})^2 & \text{for } |y-\hat{y}| \le \delta \\ \delta|y-\hat{y}| - \frac{1}{2}\delta^2 & \text{otherwise} \end{cases}$$

it is the middle of MSE & MAE.
it perform well when our data have many outliers.

4) Binary Cross Entropy

- Classification
- used when we have 2 classes in output.

$$\text{Loss fn} = -y\log(\hat{y}) - (1-y)\log(1-\hat{y})$$

$y \rightarrow$ actual value / target
$\hat{y} \rightarrow$ NN prediction

- Output activation fn Must · Sigmoid

SGS

$$\text{Cost fn} = \frac{-1}{n}\left[\sum_{i=1}^{n} y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)\right]$$

Advantage

1) Differentiable

Disadvantage

1) Multiple local minima
2) ~~Institue~~ Intuitive

- it is also called log loss.

5) Categorical Cross Entropy
Used in softmax Regression

- it is used when we have classification problem with Multi class. (class > 2)
- $L = -\sum_{j=1}^{k} y_j \log(\hat{y}_j)$ k = no. of classes in data
- Activation fn in output layer must be softmax
- $\text{Softmax} = f(z_j) = \frac{e^{z_1}}{e^{z_1}+e^{z_2}+e^{z_3}}$

output column → one hot Encoding

* Sparse Categorical Cross Entropy.
output column → integer encoding

$$C = \frac{-1}{n}\sum_{i=1}^{n}\sum_{j=1}^{k} y_{ij}\log(\hat{y}_{ij})$$

shivamkcode01